

Question:

Wayne and Bernadette are movie stars heading to the premiere of their latest film. Bernadette wants to arrive 5 minutes before Wayne. Wayne is staying at a hotel close to the premiere theater, and Bernadette is staying at her high-rise apartment in the same city. The drive from Bernadette’s apartment takes four times as long as the drive from Wayne’s hotel. If it takes Wayne four minutes to be driven to the theater, how much earlier should Bernadette leave than Wayne to get to the theater first?

Reasoning + Answer:

<think> To determine how early Bernadette should leave, we need to first calculate the time it takes her to reach the theater. Since the drive from Bernadette's apartment takes four times as long as the drive from Wayne's hotel, and it takes Wayne 4 minutes to drive to the theater, it will take Bernadette $4 \times 4 = 16$ minutes to get to the theater. Since Bernadette wants to arrive 5 minutes before Wayne, and Wayne is arriving after 4 minutes, Bernadette should arrive at $4 - 5 = -1$ minutes, which is impossible. So, this approach doesn't give the correct answer. Let's try another way. If Wayne takes 4 minutes to drive to the theater, and Bernadette takes four times as long, she takes $4 \times 4 = 16$ minutes. To arrive 5 minutes before Wayne, Bernadette should leave $16 + 5 = 21$ minutes before the time Wayne leaves. Since Wayne is going to leave after 4 minutes, Bernadette should leave after $4 - 21 = -17$ minutes, which is still not correct. The correct approach would be to find out how early Wayne needs to leave to allow Bernadette to arrive 5 minutes earlier. Wayne needs to leave enough time for Bernadette to arrive 5 minutes before him. Since Bernadette takes 16 minutes, Wayne needs to leave $16 + 5 = 21$ minutes before the time Bernadette is set to arrive. Since Bernadette is set to arrive 16 minutes after she leaves, Wayne needs to leave 21 minutes before that. So, Wayne needs to leave 21 minutes before 16, which is $16 - 21 = -5$ minutes. To find out how early Wayne needs to leave, we need the absolute value of -5, which is 5 minutes. </think><answer> 5 </answer>

